

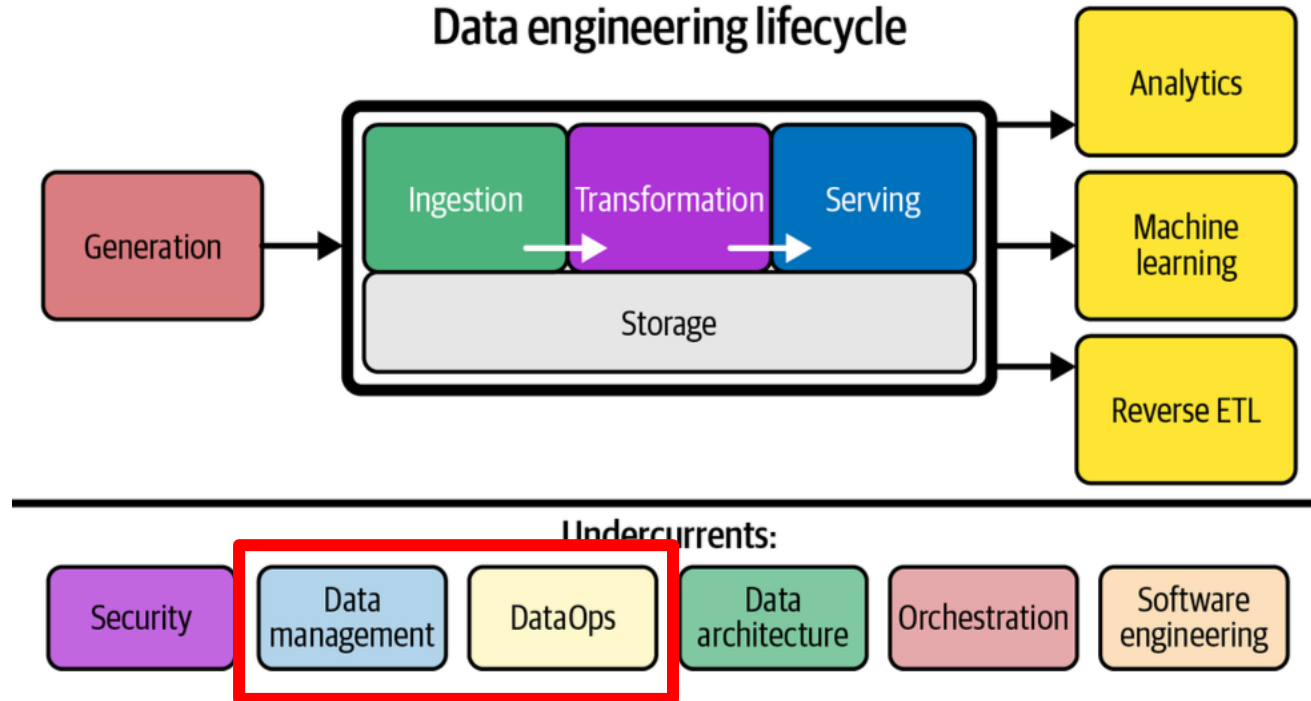
Data Engineering

Data Quality, Metadata, Lineage, and Governance

UA.DETI.ED - 2025/26

José Luís Oliveira

The Data Engineering Lifecycle

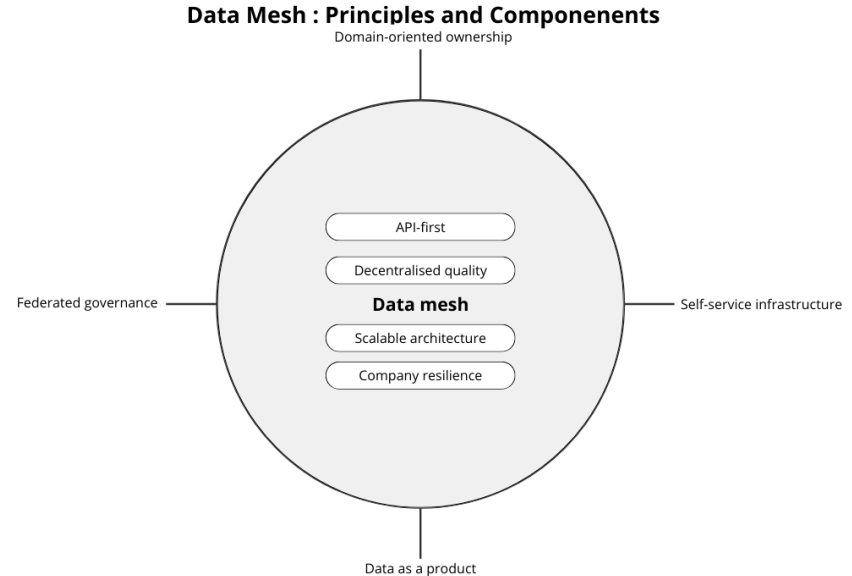


Foundations of Trustworthy Data

- ❖ Trust is the root consideration of the data engineering lifecycle.
 - Stakeholders will abandon even the most advanced platforms if the data is perceived as unreliable or inaccurate. Credibility is difficult to earn back once lost, making proactive management of data foundations essential for business ROI.
- ❖ Data Engineering is shift from "building pipelines" to "delivering data products."
 - This shift requires a focus on undercurrents such as governance, quality, and metadata management.
 - Successful data products are discoverable, trustworthy, and interoperable across the entire organization.

Data Governance

- ❖ Data governance is the framework of rules and responsibilities for managing an organization's data assets, establishing the decision rights and accountability for data-related processes and outcomes.
 - Governance ensures that data supports business strategy while minimizing risks associated with security and compliance.
- ❖ It acts as a bridge between technical implementation and business utility.
 - It formally declares data definitions to ensure everyone in the organization speaks the same language.
 - Federated governance allows decentralized teams to operate autonomously while adhering to global standards.



Data Quality

- ❖ Data quality measures how well a dataset satisfies the requirements of its intended users.
 - Quality is not absolute; it is context-dependent based on whether the data is used for real-time alerts or monthly reporting.
 - High-quality data serves as the primary fuel for accurate analytics and reliable machine learning models.
- ❖ Maintaining quality is an ongoing lifecycle process, not a one-time cleaning event.
 - It must be addressed at every stage of the data engineering lifecycle, from ingestion to serving.
 - Data observability tools are increasingly used to monitor the "health" of data systems in real-time.

Data Quality Dimension: Accuracy

- ❖ Accuracy represents the degree to which data correctly reflects the real-world objects or events it describes.
 - Inaccurate data leads to flawed insights, such as incorrect financial forecasts or misidentified customer behaviors.
 - Validating accuracy often requires comparing data against a known "golden source" or manual spot checks.
- ❖ Maintaining accuracy requires rigorous technical validation checks during the ingestion phase.
 - Automated scripts should verify that transaction amounts and timestamps match the source system's reality.
- ❖ Accuracy is often the most critical dimension for regulatory reporting and executive decision-making.

Data Quality Dimension: Completeness

- ❖ Completeness ensures that all required data is present and that there are no unexpected gaps in the dataset.
 - Missing values can significantly skew statistical averages and lead to biased machine learning training.
- ❖ Measuring completeness involves tracking the percentage of null values in critical fields over time.
- ❖ Volume tracking is another key technical method for ensuring completeness in data pipelines.
 - Sudden drops in row counts from a daily API delivery are a primary indicator of completeness issues.
- ❖ Defining "mandatory" vs. "optional" fields helps engineers prioritize validation efforts for mission-critical data.

Data Quality Dimension: Consistency

- ❖ Consistency guarantees that data values are synchronized and identical across different systems and departments.
 - Conflict arises when the Marketing team and Finance team report different numbers for the same "Revenue" metric.
 - Standardizing data formats and using shared reference data are essential for achieving consistency.
- ❖ A semantic layer can help less-technical workers gain immediate and consistent value from data.
 - Defines commonly used business metrics in terminology that can be easily and consistently understood among all users, especially non-technical users.
 - By defining business logic in a central "headless" layer, every tool in the organization retrieves the same calculated result.
 - A semantic layer abstracts the end user from the form, location, and complexity of data to deliver the following benefits.
 - Enable self-service and collaboration by business users.
 - Reduce dependency on IT and data support resources.
 - Minimize organization-wide inconsistency and duplication of effort in calculating business metrics.

Data Qual Dimensions: Validity and Timeliness

- ❖ Validity checks whether data conforms to a specific format, range, or set of business rules.
 - Valid data must adhere to predefined schemas, such as ensuring a "Zip Code" field contains exactly five digits.
 - Pattern matching and regular expressions are used in the Silver layer to filter out invalid records.
- ❖ Timeliness, or freshness, measures the delay between a real-world event and its availability in the data platform.
 - For operational analytics like fraud detection, data must be available in milliseconds to be useful.
 - Service-Level Agreements (SLAs) formally define the expected freshness for different business use cases.

Measuring Quality with SLAs and SLOs

- ❖ Service-Level Agreements (SLAs) are formal commitments between data teams and their stakeholders, and they act as a contract that sets clear expectations for data reliability across the organization.
 - They define the "what," such as a promise that data will be 99% complete and available by 8:00 AM daily.
- ❖ Service-Level Objectives (SLOs) provide the technical metrics used to measure performance against the SLA.
 - An SLO might be "95% of records must have a valid non-null customer ID during the daily ingestion run."
- ❖ Monitoring SLOs allows DataOps teams to proactively identify deteriorating quality before an SLA is breached.

Metadata Management Overview

- ❖ Metadata is the "data about data" that provides the necessary context for discovery and usage.
 - Without metadata, a data lake quickly becomes a "data swamp" where assets are impossible to find or understand.
 - Effective management requires capturing metadata automatically at every step of the engineering lifecycle.
- ❖ Metadata transforms raw tables into governed data products.
 - It enables users to understand the owner, the source, and the sensitivity of the information they are accessing.
- ❖ Metadata serves as the foundational layer for automated governance and lineage tracking.

Technical Metadata

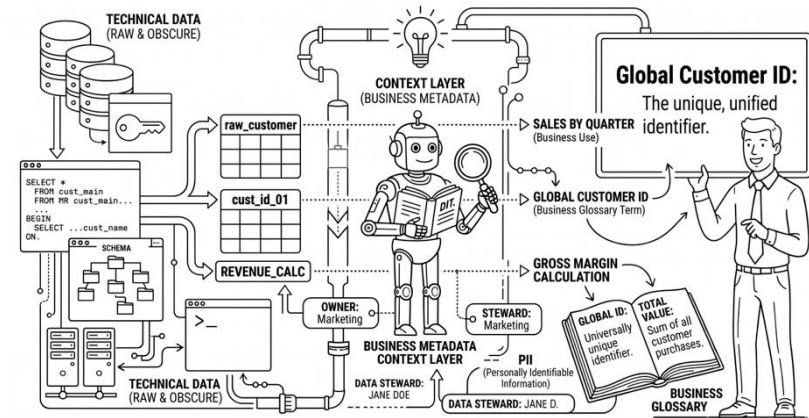
- ❖ Describes the physical structure and storage properties of the data.
 - Examples include table schemas, column names, data types, and primary key relationships.
 - This information is vital for developers and engineers building and maintaining pipelines.
- ❖ Data Structure: Defines data types (e.g., JSON, strings, integers) and schemas (fields and field types).
- ❖ Storage Details: Includes physical locations, file formats, encoding, and storage tiers.
- ❖ System Metrics: Details index parameters, database names, table definitions, and field lengths.
- ❖ Audit Traits: Tracks creation and modification timestamps.

Operational Metadata

- ❖ Operational metadata tracks the history and health of data processing jobs.
 - Examples: logs of when a job ran, how many rows were processed, and whether any errors occurred.
 - Monitoring operational metadata is the core of DataOps, enabling rapid troubleshooting and performance tuning.
- ❖ Data Lineage: Tracks the origin of a dataset and its dependencies (e.g., source systems, upstream ETL jobs, and output dashboards).
- ❖ Job Statistics: Records runtimes, execution status (success/failure), and rejected record counts during data preparation.
- ❖ Data Profiling: Displays data summaries, statistical distributions, and record volumes.
- ❖ Source Systems: Records field lengths and coding schemes originally selected from operational source systems.

Business Metadata

- ❖ Business metadata adds a layer of conceptual meaning to the raw technical data.
 - It connects technical column names like "CUST_ID_01" to business terms like "Global Customer Identifier."
- ❖ Documentation of data ownership and usage rights is a critical part of business metadata.
 - It identifies the "Data Steward" responsible for the accuracy and policy enforcement of a specific dataset.
 - Business metadata allows non-technical users to search for data using terms they understand.



The Role of Data Catalogs

- ❖ A data catalog is a centralized portal for data discovery, metadata management, and governance.
 - It acts as a "Google-like" search engine for all the data assets available within an organization.
 - Catalogs bridge the gap between technical infrastructure and the business users who need information.
- ❖ Modern catalogs use AI to automate the tagging and classification of data.
 - They can automatically detect PII (Personally Identifiable Information) to ensure compliance with privacy laws.
 - Catalogs provide a collaborative environment where users can rate datasets and share documentation.

Introduction to Data Lineage

- ❖ Data lineage is the map that shows the complete flow of data from its origin to its final destination.
 - It documents every transformation, join, and movement the data undergoes throughout its lifecycle.
 - Lineage provides the "story" of the data, allowing users to verify its authenticity and history.
- ❖ Lineage is essential for satisfying regulatory requirements like GDPR and CCPA.
 - Auditors require organizations to prove exactly how sensitive data was handled and where it is stored.
 - Lineage maps allow organizations to fulfill "right to be forgotten" requests by locating all instances of a user's data.

Technical vs. Business Lineage

- ❖ Technical lineage provides a detailed view of the physical movement of data for engineers.
 - It maps specific SQL scripts, Python functions, and table-to-table dependencies.
 - This view is critical for root cause analysis when a pipeline failure occurs.
- ❖ Business lineage provides a high-level conceptual view for analysts and stakeholders.
 - It focuses on how data moves between business domains, such as from "Sales" to "Financial Reporting."
- ❖ Business lineage ignores the technical "pipes" and focuses on the evolution of data meaning.

Impact Analysis with Lineage Tracking

- ❖ Lineage is a powerful tool for proactive change management in complex data environments.
 - Before modifying a column in a "Bronze" table, engineers can see exactly which "Gold" dashboards will break.
 - This prevents the "cascading failure" effect where one small change destroys downstream reports.
- ❖ Lineage tracking enables data teams to safely decommission obsolete datasets.
 - If the lineage map shows zero downstream consumers for a table, it can be safely deleted to reduce storage costs.
- ❖ It empowers teams to estimate the cost and effort of architectural migrations with high precision.

Troubleshooting and Root Cause Analysis

- ❖ Lineage dramatically reduces "Data Downtime" by speeding up the debugging process.
 - When a CEO sees an incorrect number on a dashboard, lineage allows engineers to trace the error back to the faulty source.
 - It answers the question: "At what point in the pipeline did this data become corrupted?"
- ❖ Automated lineage tracking eliminates the need for manual, error-prone documentation.
 - Modern tools parse SQL logs and orchestration code to generate visual maps in real-time.
 - Visual lineage graphs allow even non-technical stakeholders to understand the complexity and origin of their data.

Data Observability: The Modern Approach

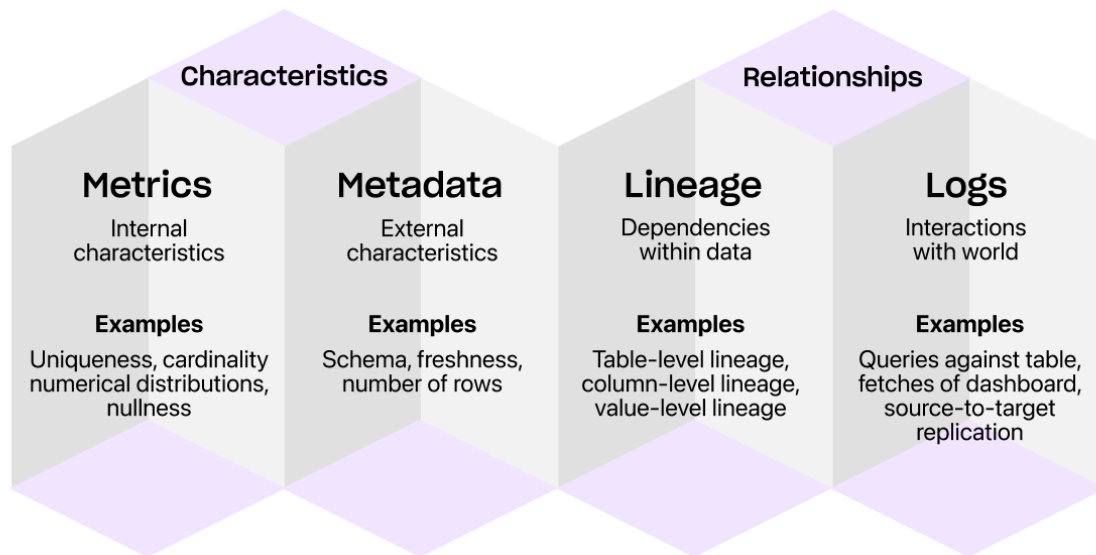
- ❖ Data observability is the evolution of data quality, focusing on proactive system health.
 - It applies the principles of software DevOps to the data engineering lifecycle.
- ❖ Observability platforms monitor for anomalies in volume, schema, and distribution without manual rule-setting.
 - It shifts the team from "reactive firefighting" to "proactive management."
 - Alerts are triggered the moment a data pattern shifts significantly from its historical baseline.
 - Combining lineage with observability allows for "automated impact alerts" sent directly to the affected users.

Data Observability Pillars

Four Pillars of Data Observability



The categories of necessary and sufficient information to describe the state of data at any point in time



<https://www.metaplane.dev/blog/the-four-pillars-of-data-observability>

AI and Automated Anomaly Detection

- ❖ Relying solely on manual, static data quality rules is virtually impossible to scale across modern enterprise data platforms.
 - Engineers cannot possibly write and maintain individual SQL assertions for every single column across tens of thousands of active tables.
 - Static thresholds also frequently trigger false positive alerts when natural business seasonality causes perfectly expected spikes or dips in data volume.
- ❖ Modern observability tools utilize machine learning to automatically establish dynamic, intelligent baselines for normal data behavior.
 - These systems analyze historical trends to deeply understand the natural rhythm of your data without requiring any manual rule configuration.
- ❖ Automated anomaly detection significantly reduces alert fatigue for busy DataOps teams.
 - By intelligently filtering out expected seasonal noise, engineers are only notified when a statistically significant and genuine data incident occurs.

Integrating Governance, Quality, and Metadata

- ❖ These three pillars are deeply interconnected and rely on each other for success.
 - Quality metrics (SLAs) are stored as metadata in the data catalog for easy visibility.
 - Lineage relies on metadata to build its map of transformations and dependencies.
- ❖ Governance provides the policies that dictate the quality standards and metadata requirements.
 - A unified strategy ensures that data is not just "available," but also compliant, accurate, and understandable.
 - Together, they build the culture of "Data Trust" required for a data-driven organization.

Further Reading & Resources

- ❖ Observability in the AI-Native Era, Hilliary Lipsig, Andreas Grabner, Robert Rati, Chapters 1-3
 - <https://learning.oreilly.com/library/view/observability-in-the/9781806389599/>
- ❖ Context Engineering for Observability Queue, David Beale
 - <https://learning.oreilly.com/library/view/context-engineering-for/0642572263003/>

